

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250275097

Kind Code

A1

Publication Date

August 28, 2025

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COOLING SYSTEMS INCLUDING STACKS WITH FINS

Abstract

Cooling systems including stacks with shark fin surface features. An example compute device including a heat-producing component, a stack thermally coupled to the heat-producing component, the stack including a first plate, a second plate, the first plate and the second plate defining a channel therebetween, a fin extending from a surface of the first plate into the channel, the fin including a first end adjacent to the surface, a second end distal to the surface, a spine extending between the first end and the second end, the spine inclined at an angle relative to the surface, and a fan to direct air through the channel.

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Family ID: 1000007919266

Appl. No.: 18/618659

Filed: March 27, 2024

Foreign Application Priority Data

WO

PCT/CN2024/078940

Feb. 28, 2024

Related U.S. Application Data

parent WO continuation PCT/CN2024/078940 20240228 PENDING child US 18618659

Publication Classification

Int. Cl.: H05K7/20 (20060101)

U.S. Cl.:

Background/Summary

FIELD OF THE DISCLOSURE

[0001] This disclosure relates generally to convection cooling systems and, more particularly, to cooling systems including stacks with fins.

BACKGROUND

[0002] Convection-based cooling systems often include a base plate, which transfers heat from a heat-producing component, such as an integrated circuit. Some such convection cooling systems include heat pipes to transfer heat from the base plate to the fin structure. Convection-based cooling systems typically include fans, which drive a flow of air over the fin structure and transfer heat from the fin structure into the ambient environment.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] FIG. 1 is a top view of an example cooling system including a stack implemented in accordance with teachings of this disclosure.

[0004] FIG. 2 is a perspective view of the stack of FIG. 1.

[0005] FIG. 3 is a perspective view of a fin of the array of fins of FIG. 2.

[0006] FIG. 4A is a schematic view of a flow through a prior fin structure.

[0007] FIG. 4B is a schematic view of an example flow through the stack of FIG. 2.

[0008] FIG. 5 is a perspective view of another fin implemented in accordance with teachings of this disclosure and that can be used in conjunction with the stack of FIG. 2.

[0009] FIG. 6 is a perspective view of another fin implemented in accordance with teachings of this disclosure and that can be used in conjunction with the stack of FIG. 2.

[0010] In general, the same reference numbers will be used throughout the drawing(s) and accompanying written description to refer to the same or like parts. The figures are not necessarily to scale.

DETAILED DESCRIPTION

[0011] Many compute devices include cooling systems to cool the heat-producing components of the compute devices. Many of these prior cooling systems are convection-based and include base plates that abut a heat-producing component of the compute system and absorb heat therefrom via conduction. Some prior convection-based cooling systems include stacks of heat conducting plates that are thermally coupled to the base plate via heat pipes. Air can be driven through the stacks via a fan, which draws air from the ambient environment. As air is driven through the stack, heat is absorbed into the air from the stack via convection. The heat exchange efficiency of a stack is a parameter that affects the thermal capacity of the convection cooling system. The heat exchange efficiency of a stack in prior convection systems is based on the number of plates of the stack, the size of the gap(s) between the plates, and/or the thickness(es) of the plates. However, the tuning of these parameters in prior cooling systems can be limited by the packaging space available to the stack, the material(s) of the plates, and/or the impedance of the flow through the stack.

[0012] Prior stacks are susceptible to recirculation and/or other turbulent flow effects in portions of the stack distal to the outlet of the fan. Recirculation of air and turbulent effects within a stack can significantly reduce the heat exchange efficiency of the stack and decrease the thermal design power (TDP) of the compute device. Another challenge posed by prior stacks is the surface drag

associated with the plates. Particularly, within the individual channels of the stacks, friction between the air and the plates causes the air near the plates to be comparatively hot and slow and air in the middle of the channel to be comparatively fast and cool. This temperature gradient can reduce the heat exchange efficiency of the stack.

[0013] Examples disclosed herein address one or more of the deficiencies described above and include convection-based cooling systems having stacks including fins based on the biomimetics of shark fins. Example fins disclosed herein include holes that enable the flow in a stack to move between adjacent channels of the stack. In some such examples disclosed herein, the holes in the fins balance pressure between adjacent channels. Example fins disclosed herein disrupt the flow of air over the surface of the fins and induce mixing between the comparatively slow hot air near the surface of the plates and the comparatively fast cool air in the middle of the channels of the stack. In some such examples disclosed herein, the mixing of air within the channels increases the heat exchange efficiency of the stack. Example fins disclosed herein mitigate recirculation effects (e.g., eddies, vortexes, whirlpools, etc.) at the inlet of the fin stacks, which increases the flow rate of air through the stacks. The example stacks disclosed herein do not have significantly increased manufacturing and/or material costs when compared to prior fin stacks. Example fin stacks disclosed herein increase the TDP of convection-based cooling systems when compared to prior convection cooling systems.

[0014] FIG. 1 is a top view of an example cooling system **100** including an example stack **102** implemented in accordance with teachings of this disclosure. In the illustrated example of FIG. 1, the cooling system **100** is disposed in an example compute device **101**. In the illustrated example of FIG. 1, the cooling system **100** includes an example base plate **104**, example heat pipes **106**, and an example fan **108**. In the illustrated example of FIG. 1, the fan **108** includes an example fan outlet **112**. In the illustrated example of FIG. 1, the stack **102** includes an example first region **114** adjacent to the fan outlet **112** and an example second region **116** distal to the fan outlet **112**.

[0015] In the illustrated example of FIG. 1, the cooling system **100** is incorporated into the compute device **101**. The cooling system **100** cools the compute device **101** and dissipates heat generated by the operation of components (e.g., processors circuitry, electronics, motors, etc.) thereof. In some examples, the compute device **101** can be a laptop, a personal computer (e.g., a tower computer, etc.), a workstation, a notebook, a tablet, a server, and/or any other suitable type of compute device. In other examples, the cooling system **100** can be incorporated into any other suitable device that includes heat-generating device(s) and/or heat-generating portions (e.g., a refrigerator unit, an air conditioner, etc.). While the example cooling system **100** described herein is an air-cooling system, in other examples, the cooling system **100** can be a liquid cooling system (e.g., an immersion cooling system, etc.). In some such examples, a liquid (e.g., a refrigerant, distilled water, etc.) can be directed by the fan **108** and/or a pump through the stack **102**.

[0016] The stack **102** of this example is a mechanical structure that includes a plurality of plates disposed in parallel. The stack **102** absorbs heat from the heat pipes **106** and dissipates the heat into the air flowing through the stack **102** and into the ambient environment. The transfer of heat from the stack **102** into the air flowing therethrough cools one or more example heat producing component(s) **118** (e.g., processor circuitry, etc.) of the compute device **101**. The stack **102** includes a plurality of fins (e.g., biomimetic shark fins shaped, etc.) that increase the heat exchange rate between the air flowing through the stack **102** and the plates of the stack **102**. For example, some or all of the plates of the stack **102** of FIG. 1 can include fins that extend between individual ones of the plates of the stack **102**. In some examples, the stack **102** includes (e.g., is composed of, etc.) a thermally conductive material (e.g., copper, aluminum, silver, etc.). An example implementation of the stack **102** is described in greater detail below in conjunction with FIG. 2.

[0017] The base plate **104** is a structural component that absorbs heat from the heat producing component(s) **118**. In the illustrated example of FIG. 1, the base plate **104** contacts (e.g., abuts, engages with, etc.) and is in thermal contact with the heat producing component(s) **118**. In some

examples, a thermally conductive paste is disposed between the heat producing component(s) **118** and the base plate **104** to increase the rate of conduction between the base plate **104** and the heat producing component(s) **118**. In some examples, the base plate **104** includes (e.g., is composed of, etc.) a thermally conductive material (e.g., copper, aluminum, etc.). In some examples, the heat producing component(s) **118** is a compute unit that performs operations for the compute device **101**. In some such examples, the heat producing component(s) **118** can be one or more central processor unit(s) (CPU), one or more graphical processor unit(s) (GPU), one or more digital signal processor(s) (DSP), one or more data processing unit(s) (DPU), one or more tensor processing unit (TPU), and/or any other suitable integrated circuitry. Additionally or alternatively, the heat producing component(s) **118** can include other component(s), such as a motor, a fan, a compressor, a bearing, a heat-exchanger, etc.

[0018] The heat pipes **106** are thermally conductive structures that transfer (e.g., conduct, etc.) heat between the base plate **104** and the stack **102**. In some examples, the heat pipes **106** include internal channels that include a fluid (e.g., ammonia, methanol, ethanol, water, mercury, etc.) that vaporizes as the fluid absorbs heat from the base plate **104**. The vapor can travel to a portion of the heat pipes **106** adjacent to the stack **102**, dissipate (e.g., conduct, transfer, etc.) heat into the stack **102**, condense into a liquid, and return to a portion of the heat pipes **106** close to the base plate **104** (e.g., via gravity, via capillary action, etc.). In some examples, the body of the heat pipes **106** can include a highly conductive material, such as copper, aluminum, silver, etc. In some examples, the internal region(s) of the heat pipes **106** have a wick/capillary design (e.g., a grooved wick design, a sintered wick design, mesh-weave wick design, etc.). In some examples, the heat pipes **106** are absent. In some examples, the base plate **104** is directly coupled to the stack **102**.

[0019] The fan **108** is a mechanical device that draws air in from the ambient environment of the compute device **101** and directs the air through the stack **102**. In the illustrated example of FIG. 1, the fan **108** is a centrifugal fan. In other examples, the fan **108** is an axial fan. In some examples, the compute device **101** can include additional fans (e.g., a fan associated with a power supply of the compute device **101**, etc.).

[0020] In the illustrated example of FIG. 1, the fan outlet **112** of the fan **108** is adjacent to the first region **114** of the stack **102** and distal to the second region **116** of the stack **102**. In some examples, the position of the fan outlet **112** can increase the relative flow rate of air through the first region **114** and decrease the relative flow rate of air through the second region **116**. In some such examples, the fins of the stack **102** can normalize (e.g., partially normalize, fully normalize, etc.) the pressure and flow rate of air between the regions **114**, **116**. In the illustrated example of FIG. 1, the air exiting the fan outlet **112** defines an example first flow vector **120** in the first region **114** and an example second flow vector **122** in the second region **116**. In some examples, the first flow vector **120** forms an example first angle **123** with an example stack flow vector **124**. In the illustrated example of FIG. 1, the second flow vector **122** forms an example second angle **126** with the stack flow vector **124**. In the illustrated example of FIG. 1, the first angle **123** is approximately **180** degrees (e.g., the first flow vector **120** is substantially colinear with the stack flow vector **124**, etc.). In the illustrated example of FIG. 1, the second angle **126** is an obtuse angle. The obtuse angle formed between the second flow vector **122** and the stack flow vector **124** causes recirculation effects within the stack **102** in the second region **116**, which is mitigated by the presence of the fins of the stack **102**. A prior stack including recirculation effects is described in conjunction in FIG. 4A. Example mitigation of such recirculation effects by example fins disclosed herein is described below in conjunction with FIG. 4B.

[0021] FIG. 2 is a perspective view of the stack **102** of FIG. 1 including a plurality of plates **200** implemented in accordance with teachings of this disclosure. In the illustrated example of FIG. 2, the plurality of plates **200** includes multiple plates such as an example first plate **202A**, an example second plate **202B**, an example third plate **202C**, etc. In the illustrated example of FIG. 2, an example first channel **204A** is defined between the first plate **202A** and the second plate **202B**. In

the illustrated example of FIG. 2, an example second channel **204B** is defined between the first plate **202A** and the third plate **202C**. In the illustrated example of FIG. 2, the second plate **202B** is depicted via a dashed line and as transparent for visuality clarity only. In the illustrated example of FIG. 2, the first plate **202A** includes an example plurality of fins **206**. In the illustrated example of FIG. 2, the stack **102** includes an example stack flow vector **124** representing the flow of air through the stack **102** (e.g., as directed from the fan **108** of FIG. 1, etc.). Although this description is focused on 3 plates (e.g., the plates **202A**, **202B**, **202C**, etc.) of the stack **102**, it is to be understood to extend to similar features of other ones of the plates **200** of the stack **102**.

[0022] The plurality of plates **200** are walls of the stack **102** that direct the flow of air through the stack **102**. As used herein, the terms “sheet” and “plate” are used interchangeably to refer to the plates of the stacks described herein. The plates **200** can include (e.g., be composed of any suitable conductive material, etc.) including copper, aluminum, silver, steel, etc. In the illustrated example of FIG. 2, the plates **200** are evenly spaced. In other examples, some or all of the plates **200** are unevenly spaced. In the illustrated example of FIG. 2, the plates **200** have a consistent thickness. In other examples, some or all of the plates **200** have different thicknesses. In some examples, the spacing between the plates **200** (e.g., the width of the first channel **204A**, etc.), the number of plates **200**, and the thickness of the plates **200** can be selected based on the design parameters of the cooling system **100**.

[0023] In the illustrated example of FIG. 2, the plates **200** include an example first lock mechanism **208A** and an example second lock mechanism **208B**. The lock mechanisms **208A**, **208B** couple the plurality of plates **200** into the stack **102**. For example, the lock mechanisms **208A**, **208B** can include one or more end features of the stacks **102** (e.g., a plurality of bosses and corresponding openings, etc.) that are joined via a plurality of interference fits. In some such examples, the stack **102** can be formed via stamping and/or other sheeting-forming manufacturing methods. For example, the plates **200** of the stack **102** can be manufactured as a series of metal sheets (e.g., copper sheets, aluminum sheets, etc.) that are coupled via the locking mechanism **208A**, **208B**. In some such examples, the plurality of fins **206** can be formed during the stamping of the plates **200**. In some such examples, the plates **200** that include the plurality of fins **206** do not have an increased material cost and/or manufacturing complexity when compared to a sheet without the plurality of fins **206**. Additionally or alternatively, the lock mechanisms **208A**, **208B** can include one or more fasteners (e.g., screws, rivets, bolts, etc.), one or more weld surfaces that are joined via a weld, one or more chemical adhesives, etc. In some examples, the lock mechanisms **208A**, **208B** are absent. In some such examples, the stack **102** is an integral component manufactured via machining and/or additive manufacturing.

[0024] The fins **206** are surface features (e.g., ridges, etc.) of the first plate **202A** that extend into the channel **204A**. The fins **206** disrupt the flow of air through the channel **204A** to increase the mixing of air within the first channel **204A** and mitigate (e.g., reduce, prevent, etc.) recirculation effects (e.g., eddies, whirlpools, vortexes, etc.) within the first channel **204A**. In some examples, the fins **206** include through-holes, which fluidly couple the channels **204A**, **204B**. While the fins **206** are depicted as a features of the first plate **202A**, it should be appreciated the other ones of the plurality of plates **200** can include fins similar to the fins **206** (e.g., the second plate **202B**, the third plate **202C**, some or all of the other plates **200**, a majority of the plates **200**, the plates **200** associated with the second region **116** of FIG. 1, etc.). For example, the third plate **202C** can include another plurality of fins that extend into the second channel **204B**.

[0025] In the illustrated example of FIG. 2, the array of fins **206** includes 28 fins. In other examples, the first plate **202A** can include a different number of fins (e.g., 1 fin, 10 fins, 20 fins, 40 fins, 100 fins, etc.). In some examples, the number of fins of the array of fins **206** can be selected based on the size of the plates **200**. In the illustrated example of FIG. 2, each fin of the array of fins **206** has a same size and/or a same shape. In the illustrated example of FIG. 3, the fins **206** are generally V-shaped ridges formed in the first plate **202A**. For example, the fins **206** are generally

shark-fin shaped (e.g., biomimetic shaped like the first dorsal fin of shark, the portion of a shark that may be seen above the water when the shark swims near the surface of the water, etc.). In other examples, some or all of the fins **206** can have different size(s) and/or different shapes. An example fin of the fins **206** is described below in conjunction with FIG. 3.

[0026] In the illustrated example of FIG. 2, the fins **206** include an example first fin **209A**, an example second fin **209B**, and an example third fin **209C**. In the illustrated example of FIG. 2, the first fin **209A** is aligned with the second fin **209B** along an example first axis **210A** parallel to the stack flow vectors **124**. In the illustrated example of FIG. 2 the first fin **209A** is aligned with the third fin **209C** along an example second axis **210B** perpendicular to the first axis **210A** and the stack flow vectors **124**. In the illustrated example of FIG. 2, the fins **206** have a grid pattern (e.g., a uniform grid distribution, etc.). That is, the fins **206** are equally distributed along the first axis **210A** and the fins **206** are equally distributed along the second axis **210B**. In other examples, the fins **206** can have another uniform distribution (e.g., adjacent rows of the fins **206** are misaligned, adjacent columns of the fins **206** are misaligned, etc.). In other examples, the fins **206** can have a non-uniform distribution (e.g., randomly distributed, unevenly spaced, etc.). In some such examples, the first fin **209A** and the second fin **209B** are unaligned along the first axis **210A** and the first fin **209A** and the third fin **209C** are unaligned along the second axis **210B**.

[0027] FIG. 3 is a perspective view of an example implementation of a fin **300** of the array of fins **206** of FIG. 2. In the illustrated example of FIG. 3, the fin **300** extends from an example surface **301** of the first plate **202A** of FIG. 2 into the channel **204A**. In the illustrated example of FIG. 3, the fin **300** includes an example first end **302**, an example second end **304**, and an example spine **306**. As used herein, the spine **306** is interchangeable referred to as a “crest” of the fin **300** and a “spine” of the fin **300**. In the illustrated example of FIG. 3, the spine **306** has an example length **308** and the second end **304** has an example height **310** and an example width **312**. In the illustrated example of FIG. 3, the fin **300** has an example thickness **314**. In the illustrated example of FIG. 3, the second end **304** includes an example hole **316**. Alternative fins implemented in accordance with teachings of this disclosure are described below in conjunction with FIGS. 5 and 6.

[0028] In the illustrated example of FIG. 3, the first end **302** is adjacent to the surface **301** and the second end **304** is distal to the surface **301**. For example, in the illustrated example of FIG. 3, the first end **302** is generally flush with the first surface **301** (e.g., the spine **306** is flush with the surface **301** at the first end **302**, etc.) and the second end **304** is spaced from the surface **301** (e.g., the spine **306** is spaced from the first surface **301** at the second end **304** by the height **310**, etc.). In this example, the spine **306** of the fin **300** that extends from the first end **302** to the second end **304** is inclined relative to the surface **301** (e.g., the spine **306** is disposed at an inclined angle relative to the surface **301**, etc.) In this example, the spine **306** terminates and/or is flush with the surface **301** of the plate **202A** at the first end **302** and slopes upward to the second end **304**. In the illustrated example of FIG. 3, the spine **306** defines an acute angle relative to the surface **301** of the plate **202A** (e.g., the angle defined between the spine **306** and the projection of the spine **306** on the surface **301**, etc.). In other examples, the first end **302** is spaced from the first surface **301**. In some such examples, the first end **302** can include a hole similar to the hole **316**.

[0029] In the illustrated example of FIG. 3, the displacement of the fin **300** increases linearly along the spine **306** between the first end **302** and the second end **304** (e.g., the slope of the spine **306** is linear between the first end **302** and the second end **304**, etc.). In other examples, the spine **306** of the fin **300** can include a curve (e.g., a logistic curve, a logarithmic curve, a quadratic curve, an exponential curve, an irregular curve, etc.) between the first end **302** and the second end **304**. In the illustrated example of FIG. 3, the displacement of the fin **300** from the surface **301** increases monotonically from the first end **302** to the second end **304**. In other examples, the displacement of the fin **300** from the surface **301** increases non-monotonically from the first end **302** and the second end **304** (e.g., spine **306** includes corrugations, etc.).

[0030] The hole **316** enables air to pass through the fin **300** and the first plate **202A**. The hole **316**

is disposed at the second end **304** and extends beneath the spine **306** into the plate **202A**. As used herein, the terms “hole” and “opening” are used interchangeable to refer to the opening in the second end **304**. That is, the hole **316** enables fluid communication through first plate **202A** and between the channels on both sides of the first plate **202A** (e.g., the channels **204A**, **204B** of FIG. **3**, etc.). In the illustrated example of FIG. **3**, the hole **316** is generally triangularly shaped (e.g., the hole **316** is a triangle, the hole **316** is V-shaped, etc.). In other examples, the hole **316** can have any other suitable shape depending on the shape of the second end **304** (e.g., polygonally shaped, circularly shaped, ovoidal shaped, C-shaped, D-shaped, etc.). In some examples, the hole **316** can extend onto the surface **301** of the first plate **202A**. In other examples, the hole **316** is absent. In some such examples, the second end **304** can be solid and/or otherwise impermeable to flow over the surface **301**. In some such examples, adjacent channels of the stack **102** are not in fluid communication.

[0031] In the illustrated example of FIG. **3**, the second end **304** includes an example edge **318**. In the illustrated example of FIG. **3**, the edge **318** includes an example first edge end **320A** and an example second edge end **320B**. In the illustrated example of FIG. **3**, the edge ends **320A**, **320B** are disposed at the surface **301**. In the illustrated example of FIG. **3**, the edge **318** is generally V-shaped. That is, the edge ends **320A**, **320B** and the spine **306** at the second end **304** form a V. In the illustrated example of FIG. **3**, the edge **318** of the second end **304** and the hole **316** are generally symmetrical about the spine **306**. For example, the portion of the edge **318** extending between the spine **306** and the first edge end **320A** is symmetrical to the portion of the edge **318** extending between the spine **306** and the second edge end **320B**. In other examples, the edge **318** and the hole **316** are asymmetrical about the spine **306**.

[0032] In the illustrated example of FIG. **3**, the spine **306** has the length **308** between the first end **302** and the second end **304**. In some examples, the length **308** of the fin **300** can be based on the length of the stack **102** (e.g., the length of the stack **102** along the first axis **210A** of FIG. **2**, etc.). In the illustrated example of FIG. **3**, the fin **300** has a consistent thickness (e.g., the thickness **314**, etc.). For example, the thickness **314** of the fin **300** can be based on the thickness of the first plate **202A**. For example, if the fin **300** is formed in the plate **202A** via stamping, the fin **300** and the first plate **202A** have an approximately same thickness. In other examples, the plate **202A** and the fin **300** can have any other suitable thickness. In the illustrated example of FIG. **3**, at the second end **304**, the spine **306** is spaced the height **310** from the surface **301**. In some examples, the width of the channels of stack **102** is approximately 1 millimeter (mm) in width. In some such examples, the height **310** can be less than or equal to 0.2 millimeters (mm). Additionally or alternatively, the height **310** can be less than or equal to one-tenth of the length **308** (e.g., the length **308** is ten times greater than the height **310**, etc.). The width **312** is the distance between the edge ends **320A**, **320B**. In some examples, the width **312** is approximately equal to twice the height **310**. In other examples, the width **312** can have any other suitable dimensions. In some examples, the length **308**, the height **310**, the width **312**, and/or the thickness **314** can be selected based on the compute device associated with the stack **102** (e.g., the compute device **101** of FIG. **1**, etc.), the size of the stack **102**, the cooling fluid (e.g., air, immersion cooling fluid, etc.), and/or the velocity of the fluid within the stack **102**.

[0033] FIG. **4A** is a schematic view of a cross-sectional schematic diagram of a channel **402** of a prior fin stack **404**. In FIG. **4A**, the channel **402** is defined between a first plate **406A** and a second plate **406B**. In the illustrated example of FIG. **4A**, the channel **402** includes a channel inlet **408** and a channel outlet **410**. A flow **412** exits a fan outlet **414** (e.g., an outlet similar to the fan outlet **112**, etc.) and enters the channel inlet **408**. In FIG. **4A**, the flow **412** is depicted as a series of discrete air particles with corresponding velocity vectors. The depiction of the flow **412** as discrete particles is not to scale and is included for visual clarity and explanatory purposes only. In FIG. **4A**, the channel **402** is not aligned with the fan outlet such that the flow **412** forms an angle **416** with the plate **406A** (e.g., an angle similar to the second angle **126** of FIG. **1**, etc.). That is, the channel **402**

is positioned in the prior fin stack **404** in a position distal to the fan outlet **414** (e.g., the channel **402** is positioned in region similar to the second region **116**, etc.).

[0034] As the flow **412** enters the channel inlet **408**, the flow **412** encounters the second plate **406B** (e.g., the downstream plate, the plate more distal to the fan outlet **414**, etc.), which increases the pressure of the flow near the second plate **406B** compared to the first plate **406A**. The pressure differential creates a vortex **418** near the first plate **406A** (e.g., the upstream plate, the plate more adjacent to the fan outlet **414**, etc.). The vortex **418** reduces the pressure recovery of the channel **402** and the overall flow rate of the flow **412** through the channel **402**. The reduced pressure recovery and flow rate through the prior channel **402** decreases the heat exchange efficiency of the prior stack **404**. Additionally, the formation of vortexes in channels of the prior stack **404** distal to the fan outlet **414** creates an even exit flow rate and exit pressure along the length of the prior stack **404**. Because drag is proportional to flow velocity squared, the uneven pressure distribution of the prior stack **404** disproportionately increases drag within the prior stack **404**, when compared to a stack with an even pressure and flow velocity distribution (e.g., stack **102** of FIG. 1, etc.).

[0035] In FIG. 4A, downstream in the channel **402**, the flow **412** assumes a flow velocity distribution **420**. The flow velocity distribution **420** is caused by surface friction (e.g., drag, viscosity force, etc.) between the flow **412** and the plates **406A**, **406B**. In the flow velocity distribution **420**, the velocity of the **412** near the center of the channel **402** is substantially faster than the velocity of the flow **412** near the plates **406A**, **406B**. Because the flow **412** absorbs heat via convection from the plates **406A**, **406B**, the portion of the flow **412** near the plates **406A**, **406B** is substantially warmer than the portion of the flow **412** in the center of the channel **402**. While some heat is transferred between the air molecules within the channel **402**, when the flow **412** exits the channel outlet **410**, the flow **412** at the center of the channel **402** can be substantially cooler than the air near the plates **406A**, **406B**. Because heat exchange rate is proportionally to the difference in temperature between the mediums exchanging heat, the comparatively greater temperature of the flow **412** near the plates **406A**, **406B** reduces the heat exchange rate of the prior stack **404** and the flow **412**.

[0036] FIG. 4B is a schematic view of a cross-sectional diagram of an example flow **422** through the channel **204A** of FIG. 2. In the illustrated example of FIG. 4B, the channel **204A** is positioned distal to the fan outlet **112** of the fan **108** of FIG. 1. That is, the channel **204A** is positioned in the second region **116** of FIG. 1. In the illustrated example of FIG. 4B, the first channel **204A** includes the first plate **202A** of FIG. 2, the third plate **202C** of FIG. 2, the first fin **209A** of FIG. 2, and the second fin **209B**. In the illustrated example of FIG. 4B, the third plate **202C** includes an example third fin **424A** and an example fourth fin **424B**. In the illustrated example of FIG. 4B, the flow **422** enters the channel **204A** and forms the second angle **126** of FIG. 1 with the plates **202A**, **202B**. In the illustrated example of FIG. 4B, the channel **204A** has an example channel inlet **428** and an example channel outlet **430**.

[0037] In the illustrated example of FIG. 4B, the fins **209A**, **209B** extend from the first plate **202A** into the channel **204A**. In the illustrated example FIG. 4B, the first plate **202A** is upstream of the second plate **202B** relative to the fan outlet **112**. That is, the fins **209A**, **209B** extend from the downstream plate (e.g., the plate closer to the fan outlet **112** of the fan **108**, etc.) into the channel **204A**. In the illustrated example of FIG. 4B, the ends of the fins **209A**, **209B** that extend from the surface **301** of the first plate **202A** (e.g., the ends similar to the first end **302** of FIG. 3, etc.) are positioned closer to the channel inlet **428** than the channel outlet **430**. Similarly, the ends of the fins **209A**, **209B** that are displaced from the surface **301** of the first plate **202A** (e.g., the ends similar to the second end **304** of the FIG. 3, etc.) are positioned closer to the channel outlet **430** than the channel inlet **428**. In some such examples, the orientation of the fins **209A**, **209B** causes the portion of the flow **422** near the surface **301** within the first channel **204A** to move along the surface of the fins **209A**, **209B** and mix with the portion of the flow near the center of the first channel **204A**. The mixing of the flow **422** causes a more uniform temperature distribution of the flow **422** within the

first channel **204A**. That is, the fins **209A**, **209B** facilitate the mixing of the comparatively slow hot portion of the flow **422** near the surface **301** of the first plate **202A** and an example second surface **432** of the second plate **202B** with the comparatively fast and cold portion of the flow **422** in the center of the first channel **204A**. The mixing of the flow **422** increases the temperature differential between the portion of the flow **422** near the surfaces **301**, **432** and the plates **202A**, **202C**, which increases the heat transfer rate (e.g., the rate of convection, etc.) therebetween.

[0038] In the illustrated example of FIG. **4B**, the fins **209A**, **209B**, **424A**, **424B** include an example first hole **434A**, an example second hole **434B**, an example third hole **434C**, and an example fourth hole **434D**, respectively. The holes **434A**, **434B**, **434C**, **434D** are similar to the hole **316** of FIG. **3**. In the illustrated example of FIG. **4B**, the holes **434A**, **434B** enable fluid communication between the first channel **204A** and the second channel **204B**. In the illustrated example of FIG. **4B**, the holes **434C**, **434D** enable fluid communication between the first channel **204A** and an example third channel **436** formed between the third plate **202C** and an adjacent plate of the stack **102**. That is, the holes **434A**, **434B**, **434C**, **434D** enable some of the flow **422** to move from the first channel **204A** into the second channel **204B** and/or the third channel **436**. The movement of the flow **422** between the channels **204A**, **204B**, **436** normalizes (e.g., equalizes, etc.) the pressure between the channels **204A**, **204B**, **436**. The normalization of pressure between the channels **204A**, **204B**, **436** normalizes the flow velocity in the channels **204A**, **204B**, **436**, which reduces the overall drag associated with the flow of air through the stack **102** when compared to the flow through the prior stack **404** of FIG. **4A**. In some examples, the holes **434A**, **434B**, **434C**, **434D** create a negative pressure zone near the holes **434A**, **434B**, **434C**, **434D** due to the Bernoulli effect, which accelerates the flow **422** away from the surfaces **301**, **432** and further facilitates the mixing of the flow **422** within the first channel **204A**.

[0039] The reduced drag associated with the stack **102** decreases the power demand of the fan **108** and/or reduces the energy of the flow **422** lost to surface friction, drag, and/or viscous effects. In some such examples, the reduction in energy losses of the flow **422** increases the overall flow rate of the flow **422** within the stack **102**, which increases the heat transfer rate (e.g., the rate of convection, etc.) between the stack **102** and the flow **422**. Additionally or alternatively, the reduction in energy losses of the flow **422** enables a smaller and/or less powerful fan to be used in the cooling system **100** when compared to a cooling system including the prior stack **404** of FIG. **4A**.

[0040] In some examples, the shape of the fins **209A**, **209B**, **424A**, **424B** increases the area of the plates **202A**, **202B** exposed to the flow **422**. For example, the presence of the fins **209A**, **209B**, **424A**, **424B** can increase the area of the plates **202A**, **202B** exposed to the flow **422** by 5% when compared to the prior plates **406A**, **406B** of FIG. **4A**. In some such examples, the increased area of the fins **209A**, **209B** increases the heat exchange rate between the stack **102** and the flow **422** when compared to the prior stack **404** of FIG. **4A**. Additionally, the fins **209A**, **209B** disrupt the flow **422** near the channel inlet **428**, which disrupts the formation of vortices within the channel **204A**. In some examples, the normalization of the pressure and/or the flow rate between the channels **204A**, **204B**, **436** associated with the holes **434A**, **434B**, **434C**, **434D** further reduces (e.g., retards, prevents, mitigates, etc.) recirculation and other turbulent flow effects within the channel **204A** (e.g., vortexes, whirlpools, eddies, etc.). As such, the fins **209A**, **209B**, **424A**, **424B** and the holes **434A**, **434B**, **434C**, **434D** mitigate (e.g., reduce, prevent, etc.) the formation of vortexes within the channel **204A** and the associated pressure and velocity reduction.

[0041] FIG. **5** is a perspective view of another example fin **500** implemented in accordance with teachings of this disclosure and that can be used in conjunction with the stack **102** of FIG. **2**. The fin **500** is similar to the fin **300** of FIG. **3**, except as noted otherwise. In some examples, the fin **500** can implement one or more of the array of fins **206** of FIG. **2** (e.g., one or more of the fins **209A**, **209B**, **209C**, etc.). In the illustrated example of FIG. **5**, the fin **500** includes an example first end **502**, an example second end **504**, and an example crest **506**, which are similar to the first end **302**,

the second end **304**, and the spine **306**, respectively. In the illustrated example of FIG. 5, the second end **504** includes an example edge **508**, which includes an example first edge end **510A** and an example second edge end **510B**, which are similar to first edge end **320A** and the second edge end **320B**, respectively, except as noted otherwise. In the illustrated example of FIG. 5, the edge **508** includes an example crest edge **512**. In the illustrated example of FIG. 5, the fin **500** includes an example hole **514**, which is similar to the hole **316**, respectively.

[0042] In the illustrated example of FIG. 5, the crest edge **512** is disposed closer toward the first end **502** than the edge ends **510A**, **510B**. That is, the crest edge **512** is inset toward the first end **502**. In the illustrated example of FIG. 5, the inseting of the crest edge **512** increases the relative size of the hole **514**. In other examples, the crest edge **512** can be displaced away from the first end **502**. That is, the crest edge **512** can be displaced forward of the second end **504**, such that crest edge **506** is upstream of the edge ends **510A**, **510B**. Similarly to the fin **300** of FIG. 3, the fin **500** is generally V-shaped. The fin **500** has a comparatively deeper (e.g., stronger, etc.) V-shape than the fin **300** of FIG. 3. The comparatively larger size of the hole **514** increases the flow rate between adjacent channels of the stack **102** when compared to the hole **316** of the fin **300** of FIG. 3.

[0043] FIG. 6 is a perspective view of another example fin **600** implemented in accordance with teachings of this disclosure and that can be used in conjunction with the stack **102** of FIG. 2. The fin **600** is similar to the fin **300** of FIG. 3, except as noted otherwise. In some examples, the fin **600** can implement one or more of the array of fins **206** of FIG. 2 (e.g., one or more of the fins **209A**, **209B**, **209C**, etc.). In the illustrated example of FIG. 6, the fin **600** includes an example first end **602**, an example second end **604**, and an example crest **606**, which are similar to the first end **302**, the second end **304**, and the spine **306**, respectively, except as noted otherwise. In the illustrated example of FIG. 6, the second end **604** includes an example edge **608**. In the illustrated example of FIG. 6, the fin **600** includes an example hole **612**, which is similar to the hole **316** of FIG. 3. In the illustrated example of FIG. 6, the hole **612** includes an example first hole portion **614A** in the second end **604** and an example second hole portion **614B** on the surface **301**. As used herein, the hole portions **614A**, **614B** are also referred to interchangeably as “holes” (e.g., the first hole portion **614A** is a first hole, the second hole portion **614B** is a second hole, etc.), openings (e.g., the first hole portion **614A** is a first opening, the second hole portion **614B** is a second opening, etc.), and opening portions (e.g., the first hole portion **614A** is a first opening portion, the second hole portion **614B** is a second opening portions, etc.). In the illustrated example of FIG. 6, the first hole portion **614A** and the second hole portion **614B** are a single hole (e.g., the hole portions **614A**, **614B** are continuous, etc.). In other examples, the hole portions **614A**, **614B** are discrete holes (e.g., the a portion of the plate **202A** separates the hole portions **614A**, **614B**, etc.). In the illustrated example of FIG. 6, the edge **608** extends around both of the hole portions **614A**, **614B**. In the illustrated example of FIG. 6, the hole portions **614A**, **614B** are separated by an example first junction **616A** and an example second junction **616B**. In the illustrated example of FIG. 6, the junctions **616A**, **616B** are disposed at the interface between the second end **604** and the surface **301**. In the illustrated example of FIG. 6, the junctions **616A**, **616B** define an obtuse angle between the surface **301** and the second end **604**. In other examples, the junctions **616A**, **616B** can define a right angle and/or an acute angle.

[0044] “Including” and “comprising” (and all forms and tenses thereof) are used herein to be open ended terms. Thus, whenever a claim employs any form of “include” or “comprise” (e.g., comprises, includes, comprising, including, having, etc.) as a preamble or within a claim recitation of any kind, it is to be understood that additional elements, terms, etc., may be present without falling outside the scope of the corresponding claim or recitation. As used herein, when the phrase “at least” is used as the transition term in, for example, a preamble of a claim, it is open-ended in the same manner as the term “comprising” and “including” are open ended. The term “and/or” when used, for example, in a form such as A, B, and/or C refers to any combination or subset of A, B, C such as (1) A alone, (2) B alone, (3) C alone, (4) A with B, (5) A with C, (6) B with C, or (7) A

with B and with C. As used herein in the context of describing structures, components, items, objects and/or things, the phrase “at least one of A and B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B. Similarly, as used herein in the context of describing structures, components, items, objects and/or things, the phrase “at least one of A or B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B. As used herein in the context of describing the performance or execution of processes, instructions, actions, activities, etc., the phrase “at least one of A and B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B. Similarly, as used herein in the context of describing the performance or execution of processes, instructions, actions, activities, etc., the phrase “at least one of A or B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B.

[0045] As used herein, singular references (e.g., “a”, “an”, “first”, “second”, etc.) do not exclude a plurality. The term “a” or “an” object, as used herein, refers to one or more of that object. The terms “a” (or “an”), “one or more”, and “at least one” are used interchangeably herein.

Furthermore, although individually listed, a plurality of means, elements, or actions may be implemented by, e.g., the same entity or object. Additionally, although individual features may be included in different examples or claims, these may possibly be combined, and the inclusion in different examples or claims does not imply that a combination of features is not feasible and/or advantageous.

[0046] As used herein, unless otherwise stated, the term “above” describes the relationship of two parts relative to Earth. A first part is above a second part, if the second part has at least one part between Earth and the first part. Likewise, as used herein, a first part is “below” a second part when the first part is closer to the Earth than the second part. As noted above, a first part can be above or below a second part with one or more of: other parts therebetween, without other parts therebetween, with the first and second parts touching, or without the first and second parts being in direct contact with one another.

[0047] As used in this patent, stating that any part (e.g., a layer, film, area, region, or plate) is in any way on (e.g., positioned on, located on, disposed on, or formed on, etc.) another part, indicates that the referenced part is either in contact with the other part, or that the referenced part is above the other part with one or more intermediate part(s) located therebetween.

[0048] As used herein, connection references (e.g., attached, coupled, connected, and joined) may include intermediate members between the elements referenced by the connection reference and/or relative movement between those elements unless otherwise indicated. As such, connection references do not necessarily infer that two elements are directly connected and/or in fixed relation to each other. As used herein, stating that any part is in “contact” with another part is defined to mean that there is no intermediate part between the two parts.

[0049] Unless specifically stated otherwise, descriptors such as “first,” “second,” “third,” etc., are used herein without imputing or otherwise indicating any meaning of priority, physical order, arrangement in a list, and/or ordering in any way, but are merely used as labels and/or arbitrary names to distinguish elements for ease of understanding the disclosed examples. In some examples, the descriptor “first” may be used to refer to an element in the detailed description, while the same element may be referred to in a claim with a different descriptor such as “second” or “third.” In such instances, it should be understood that such descriptors are used merely for identifying those elements distinctly within the context of the discussion (e.g., within a claim) in which the elements might, for example, otherwise share a same name.

[0050] As used herein, “approximately” and “about” modify their subjects/values to recognize the potential presence of variations that occur in real world applications. For example, “approximately” and “about” may modify dimensions that may not be exact due to manufacturing tolerances and/or other real world imperfections as will be understood by persons of ordinary skill in the art. For

example, “approximately” and “about” may indicate such dimensions may be within a tolerance range of $\pm 10\%$ unless otherwise specified herein.

[0051] As used herein “substantially real time” refers to occurrence in a near instantaneous manner recognizing there may be real world delays for computing time, transmission, etc. Thus, unless otherwise specified, “substantially real time” refers to real time +1 second.

[0052] As used herein, the phrase “in communication,” including variations thereof, encompasses direct communication and/or indirect communication through one or more intermediary components, and does not require direct physical (e.g., wired) communication and/or constant communication, but rather additionally includes selective communication at periodic intervals, scheduled intervals, aperiodic intervals, and/or one-time events.

[0053] As used herein, “programmable circuitry” is defined to include (i) one or more special purpose electrical circuits (e.g., an application specific circuit (ASIC)) structured to perform specific operation(s) and including one or more semiconductor-based logic devices (e.g., electrical hardware implemented by one or more transistors), and/or (ii) one or more general purpose semiconductor-based electrical circuits programmable with instructions to perform specific functions(s) and/or operation(s) and including one or more semiconductor-based logic devices (e.g., electrical hardware implemented by one or more transistors). Examples of programmable circuitry include programmable microprocessors such as Central Processor Units (CPUs) that may execute first instructions to perform one or more operations and/or functions, Field Programmable Gate Arrays (FPGAs) that may be programmed with second instructions to cause configuration and/or structuring of the FPGAs to instantiate one or more operations and/or functions corresponding to the first instructions, Graphics Processor Units (GPUs) that may execute first instructions to perform one or more operations and/or functions, Digital Signal Processors (DSPs) that may execute first instructions to perform one or more operations and/or functions, XPU, Network Processing Units (NPUs) one or more microcontrollers that may execute first instructions to perform one or more operations and/or functions and/or integrated circuits such as Application Specific Integrated Circuits (ASICs). For example, an XPU may be implemented by a heterogeneous computing system including multiple types of programmable circuitry (e.g., one or more FPGAs, one or more CPUs, one or more GPUs, one or more NPUs, one or more DSPs, etc., and/or any combination(s) thereof), and orchestration technology (e.g., application programming interface(s) (API(s)) that may assign computing task(s) to whichever one(s) of the multiple types of programmable circuitry is/are suited and available to perform the computing task(s).

[0054] As used herein integrated circuit/circuitry is defined as one or more semiconductor packages containing one or more circuit elements such as transistors, capacitors, inductors, resistors, current paths, diodes, etc. For example an integrated circuit may be implemented as one or more of an ASIC, an FPGA, a chip, a microchip, programmable circuitry, a semiconductor substrate coupling multiple circuit elements, a system on chip (SoC), etc.

[0055] Examples disclosed herein include stacks with 3D shark fin perforations that can be used in conjunction with prior fin designs. Examples disclosed herein can be used in conjunction with any cooling systems that include fin structures (e.g., any device that includes a fan and stack, etc.). For example, the fin structures disclosed herein can be used in conjunction with notebook computers, desktops, and workstations. Examples disclosed herein increase the performance (e.g., the efficiency, the cooling capacity, etc.) of air-cooled computing systems. Examples disclosed herein increase the thermal capacity and thermal design power of cooling systems including such stacks. Some example stacks disclosed herein increase the thermal design power (TDP) of compute devices by approximately 7%. Examples disclosed herein do not increase the material cost or manufacturing time of stacks when compared to prior fin designs.

[0056] Cooling systems including stacks with fins are disclosed herein. Further examples and combinations thereof include the following:

[0057] Example 1 includes a compute device including a heat-producing component, a stack

thermally coupled to the heat-producing component, the stack including a first plate, a second plate, the first plate and the second plate defining a channel therebetween, a fin extending from a surface of the first plate into the channel, the fin including a first end adjacent to the surface, a second end distal to the surface, a spine extending between the first end and the second end, the spine inclined at an angle relative to the surface, and a fan to direct air through the channel.

[0058] Example 2 includes the compute device of any preceding example, wherein the channel includes an inlet adjacent to the fan, and an outlet adjacent to the fan, the first end closer to the inlet than the outlet.

[0059] Example 3 includes the compute device of any preceding example, wherein the first plate is adjacent to the fan and the second plate is distal to the fan.

[0060] Example 4 includes the compute device of any preceding example, wherein the fin further includes an opening at the second end.

[0061] Example 5 includes the compute device of any preceding example, wherein the opening is triangular.

[0062] Example 6 includes the compute device of any preceding example, wherein the opening includes a first opening portion in the fin, and a second opening portion on the surface.

[0063] Example 7 includes the compute device of any preceding example, wherein the first end is flush with the surface and the second end is displaced from the surface.

[0064] Example 8 includes the compute device of any preceding example, wherein the fin is a first fin, the stack further including a second fin extending from the first plate into the channel, the second fin aligned with the first fin along a first axis, and a third fin extending from the first plate into the channel, the third fin aligned with the first fin along a second axis perpendicular to the first axis.

[0065] Example 9 includes the compute device of any preceding example, wherein the fin is a first fin, the channel is a first channel and the stack further includes a third plate, the first plate between the second plate and the third plate, the third plate and the first plate defining a second channel, and a second fin extending from the third plate into the second channel.

[0066] Example 10 includes the compute device of any preceding example, wherein the spine extends monotonically between the first end and the second end.

[0067] Example 11 includes the compute device of any preceding example, wherein the spine extends linearly between the first end and the second end.

[0068] Example 12 includes a thermally conductive stack comprising a first plate, a second plate, the first plate and the second plate defining a channel, and a fin extending from a surface of the first plate into the channel, the fin including a first end adjacent to the surface, a second end distal to the surface, and a spine extending between the first end and the second end, the spine disposed at an incline angle relative to the surface.

[0069] Example 13 includes the thermally conductive stack of any preceding example, wherein the second end includes a V-shaped opening.

[0070] Example 14 includes the thermally conductive stack of any preceding example, wherein the V-shaped opening extends through the first plate.

[0071] Example 15 includes the thermally conductive stack of any preceding example, wherein the V-shaped opening includes a first opening in the second end, and a second opening in the surface.

[0072] Example 16 includes the thermally conductive stack of any preceding example, wherein the first plate further includes a plurality of fins, the plurality of fins including the fin.

[0073] Example 17 includes the thermally conductive stack of any preceding example, wherein the plurality of fins have a grid pattern on the first plate.

[0074] Example 18 includes the thermally conductive stack of any preceding example, wherein the fin is a first fin, the channel is a first channel and the thermally conductive stack further includes a third plate, the first plate between the second plate and the third plate, the third plate and the second plate defining a second channel, and a second fin extending into the second channel.

[0075] Example 19 includes the thermally conductive stack of any preceding example, wherein the fin increases a mixing of a flow in the channel.

[0076] Example 20 includes the thermally conductive stack of any preceding example, wherein the spine extends monotonically between the first end and the second end.

[0077] Example 21 includes the thermally conductive stack of any preceding example, wherein the spine extends linearly between the first end and the second end.

[0078] Example 22 includes the thermally conductive stack of any preceding example, wherein the spine defines an acute angle with the surface of the first plate.

[0079] Example 23 includes a thermally conductive sheet of a cooling system, the thermally conductive sheet comprising a first fin extending from a surface of the thermally conductive sheet, the first fin including a first end flush with the surface, a second end displaced from the surface, and a spine extending between the first end and the second end, and a second fin extending from the surface having a same shape as the first fin.

[0080] Example 24 includes the thermally conductive sheet of any preceding example, wherein the first fin includes an edge at the second end, the edge including a third end at the surface, and a fourth end at the surface, the spine inset toward the first end relative to the third end and the fourth end.

[0081] Example 25 includes the thermally conductive sheet of any preceding example, wherein the second end is spaced a height from the thermally conductive sheet, the spine having a length, and the spine is at least ten times greater than the height.

[0082] Example 26 includes the thermally conductive sheet of any preceding example, wherein the second end includes an opening extending through the thermally conductive sheet.

[0083] Example 27 includes the thermally conductive sheet of any preceding example, wherein the opening includes an edge, the edge including a junction having an obtuse angle.

[0084] Example 28 includes the thermally conductive sheet of any preceding example, wherein the thermally conductive sheet has a first thickness, and the first fin has a second thickness substantially equal to the first thickness.

[0085] Example 29 includes the thermally conductive sheet of any preceding example, wherein the first fin is shark-fin shaped.

[0086] Example 30 includes the thermally conductive sheet of any preceding example, wherein the spine extends monotonically between the first end and the second end.

[0087] Example 31 includes the thermally conductive sheet of any preceding example, wherein the spine defines an acute angle with the surface.

[0088] The following claims are hereby incorporated into this Detailed Description by this reference. Although certain example systems, apparatus, articles of manufacture, and methods have been disclosed herein, the scope of coverage of this patent is not limited thereto. On the contrary, this patent covers all systems, apparatus, articles of manufacture, and methods fairly falling within the scope of the claims of this patent.

Claims

1. A compute device including: a heat-producing component; a stack thermally coupled to the heat-producing component, the stack including: a first plate; a second plate, the first plate and the second plate defining a channel therebetween; a fin extending from a surface of the first plate into the channel, the fin including: a first end adjacent to the surface; a second end distal to the surface; a spine extending between the first end and the second end, the spine inclined at an angle relative to the surface; and a fan to direct air through the channel.
2. The compute device of claim 1, wherein the channel includes: an inlet adjacent to the fan; and an outlet adjacent to the fan, the first end closer to the inlet than the outlet.
3. The compute device of claim 2, wherein the first plate is adjacent to the fan and the second plate

is distal to the fan.

4. The compute device of claim 1, wherein the fin further includes an opening at the second end.

5. The compute device of claim 4, wherein the opening includes: a first opening portion in the fin; and a second opening portion on the surface.

6. The compute device of claim 1, wherein the first end is flush with the surface and the second end is displaced from the surface.

7. The compute device of claim 1, wherein the fin is a first fin, the channel is a first channel and the stack further includes: a third plate, the first plate between the second plate and the third plate, the third plate and the first plate defining a second channel; and a second fin extending from the third plate into the second channel.

8. The compute device of claim 1, wherein the spine extends linearly between the first end and the second end.

9. A thermally conductive stack comprising: a first plate; a second plate, the first plate and the second plate defining a channel; and a fin extending from a surface of the first plate into the channel, the fin including: a first end adjacent to the surface; a second end distal to the surface; and a spine extending between the first end and the second end, the spine disposed at an incline angle relative to the surface.

10. The thermally conductive stack of claim 9, wherein the second end includes a V-shaped opening.

11. The thermally conductive stack of claim 10, wherein the V-shaped opening extends through the first plate.

12. The thermally conductive stack of claim 11, wherein the V-shaped opening includes: a first opening in the second end; and a second opening in the surface.

13. The thermally conductive stack of claim 9, wherein the first plate further includes a plurality of fins, the plurality of fins including the fin.

14. The thermally conductive stack of claim 9, wherein the fin is a first fin, the channel is a first channel and the thermally conductive stack further includes: a third plate, the first plate between the second plate and the third plate, the third plate and the second plate defining a second channel; and a second fin extending into the second channel.

15. The thermally conductive stack of claim 9, wherein the fin increases a mixing of a flow in the channel.

16. The thermally conductive stack of claim 9, wherein the spine extends monotonically between the first end and the second end.

17. A thermally conductive sheet of a cooling system, the thermally conductive sheet comprising: a first fin extending from a surface of the thermally conductive sheet, the first fin including: a first end flush with the surface; a second end displaced from the surface; and a spine extending between the first end and the second end; and a second fin extending from the surface having a same shape as the first fin.

18. The thermally conductive sheet of claim 17, wherein the first fin includes an edge at the second end, the edge including: a third end at the surface; and a fourth end at the surface, the spine inset toward the first end relative to the third end and the fourth end.

19. The thermally conductive sheet of claim 17, wherein the second end includes an opening extending through the thermally conductive sheet.

20. The thermally conductive sheet of claim 17, wherein the spine extends monotonically between the first end and the second end.
